

Behavioral dataset for Long-Evans and its schizophrenia-like substrain through several generations

Feature descriptor tables

This document contains useful information before using the published datasets. Here one can find the feature descriptions for both raw and processed datasets.

Table 1: Feature description for LE_Lisket_0_15_AMBI_DEF_2.xlsx

Feature name	Description	Unit	Note
Metadata			
Animal	Animal ID	Text (str)	
Generation	Number of the generation where the rat belongs	Number (int or float)	
Season	The season when the animals were delivered	Date	
Separation	Date of the separation from the bunch	Date	
G_S	The season of the different generations	Text (str)	
Paradigm	Types of tasks: Task1 or Task2	Text (str)	
Date_Ambitus	Date of Ambitus test	Date	
Year	Year of the experiment (0-6)	Number (int or float)	
NR	ID number of the animal	Number (int or float)	
Group	The group label of the rat	Text (str)	
Sex	Sex of the rat		Text (str)
GR_Sex	The sex of the animals in the different groups together	Text (str)	
Trials	Number of trial	Number (int or float)	
id	Unique identifier constructed from Group, NR and Generation	Text (str)	
Animal	Identifier of the animal in a generation	Text (str)	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Reward collection related parameters			
EAT_E_Nr	The number of the collected rewards from the external boxes	Number (int or float)	Only trial 1-2
EAT_E_%	The percentage of the collected rewards from the external boxes (number of collected rewards x 100) / (available rewards [8])	%	Only trial 1-2
EAT_I_Nr	The number of the collected rewards from the internal boxes	Number (int or float)	
EAT_I_%	The percentage of the collected rewards from the internal boxes (number of collected rewards x 100) / (available rewards [8])	%	
EAT_TOT_Nr	The number of the collected rewards from all boxes	Number (int or float)	
EAT_TOT_%%	The percentage of the collected rewards from all boxes (number of collected rewards x 100) / (available rewards [16])	%	
EAT_E_T	The time up to the collection of the rewards from the external boxes (cut off time: 300s)	Second	Only trial 1-2
EAT_I_T	The time up to the collection of the rewards from the internal boxes (cut off time: 300s)	Second	
EAT_TOT_T	The time up to the collection of the rewards from all boxes (cut off time: 300s)	Second	
Locomotion related parameters			
Dir_Change_Nr	The number of direction changes during the 300s	Number (int or float)	
Circle_Nr	The number of total circles during the 300s	Number (int or float)	
LOCO_TOT	The number of entries into the corridors during the 300s	Number (int or float)	
Explored Area	The number of visited corridors (max:4)	Number (int or float)	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Loco_BEf_NR	The number of entries into the corridors up to collection of rewards (Number of corridor entries up to collection of rewards $\times$ 300) / EAT_TOT_T	Number (int or float)	
LOCO_BEf	Locomotion frequency up to the task completion. (Number of entries in a corridor $\times$ 300) EAT_TOT_T	Number (int or float)	
Exploratory activity related parameters			
Expl_E_BEf_Nr	Number of external box visits before reward collection.	Number (int or float)	
Expl_I_BEf_Nr	Number of internal box visits before reward collection.	Number (int or float)	
Expl_E_I_BEf_Nr	Number of all box visits before reward collection.	Number (int or float)	
Expl_E_BEf_Calc	Subset exploration categorized as external box visits up to collection of the rewards (Number of external box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_I_BEf_Calc	Subset of exploration categorized as internal box visits up to collection of the rewards (Number of internal box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	
EXPL_BEf	Subset of exploration up to collection of the rewards (Number of box exploration up to collection of reward $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_E_AFT_Nr	Number of external box exploration after the collection of the rewards	Number (int or float)	If the eating time is 300, there is no value
Expl_I_AFT_Nr	Number of internal box exploration after the collection of the rewards	Number (int or float)	If the eating time is 300, there is no value
Expl_E_I_AFT_Nr	Number of all box exploration after the collection of the rewards	Number (int or float)	If the eating time is 300, there is no value

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Expl_E_AFT_Calc	Subset of external box exploration after the collection of all rewards (Number of external box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	If the eating time is 300, there is no value
Expl_I_AFT_Calc	Subset of internal exploration after the collection of all rewards (Number of internal box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	If the eating time is 300, there is no value
Expl_E_I_AFT_Calc	Subset of exploration after the collection of all rewards (Number of box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	If the eating time is 300, there is no value
Expl_E_BEF_AFT_Nr	Number of external box exploration during 300s	Number (int or float)	
Expl_I_BEF_AFT_Nr	Number of internal box exploration during 300s	Number (int or float)	
EXPL_TOT	Number of all box exploration during 300s	Number (int or float)	
Expl_E_BEF_T	Duration of external box visits up to the collection of rewards	Second	
Expl_I_BEF_T	Duration of internal box visits up to the collection of rewards	Second	
Expl_E_I_BEF_T	Duration of box visits up to the collection of rewards	Second	
Expl_E_BEF_Calc_T	Subset exploration time categorized as external box visits up to collection of the rewards (Duration of external box exploration $\times$ 300) / EAT_TOT_T	Second	
Expl_I_BEF_Calc_T	Subset exploration time categorized as external box visits up to collection of the rewards (Duration of external box exploration $\times$ 300) / EAT_TOT_T	Second	
Expl_E_I_BEF_Calc_T	Subset exploration time categorized as external box visits up to collection of the rewards (Duration of external box exploration $\times$ 300) / EAT_TOT_T	Second	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Expl_E_AFT_T	Duration of external box visits after the collection of all rewards	Second	If the eating time is 300, there is no value
Expl_I_AFT_T	Duration of internal box visits after the collection of all rewards	Second	If the eating time is 300, there is no value
Expl_E_I_AFT_T	Duration of box visits after the collection of all rewards	Second	If the eating time is 300, there is no value
Expl_E_AFT_T_Calc	Subset exploration time categorized as external box visits after the collection of all rewards (Duration of external box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Second	If the eating time is 300, there is no value
Expl_I_AFT_T_Calc	Subset exploration time categorized as internal box visits after the collection of all rewards (Duration of internal box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Second	If the eating time is 300, there is no value
Expl_E_I_AFT_T_Calc	Subset exploration time categorized a box visits after the collection of all rewards (Duration of box exploration after the collection of all rewards $\times$ 300) / (300-EAT_TOT_T)	Second	If the eating time is 300, there is no value
Expl_E_BEF_AFT_T	Duration of external box exploration during 300s	Second	
Expl_I_BEF_AFT_T	Duration of internal box exploration during 300s	Second	
Expl_TOT_T	Duration of all box exploration during 300s	Second	
Expl_E_BEF_Loco_ratio	Locomotion-independent exploration into the external boxes up to the EAT_TOT_T: The ratio of external visits to locomotor activity until all rewards were collected	Number (int or float)	Only trial 1-2
Expl_I_BEF_Loco_ratio	Locomotion-independent exploration into the internal boxes up to the EAT_TOT_T: The ratio of internal visits to locomotor activity during the 300s	Ratio	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Expl_EI_BEF_Loco_ratio	Locomotion-independent exploration into the boxes up to the EAT_TOT_T: The ratio of visits to locomotor activity during the 300s	Ratio	
Expl_E_TOT_Loco_ratio	Locomotion-independent exploration into the external boxes during the 300s: The ratio of external visits to locomotor activity during the 300s	Ratio	
Expl_I_TOT_Loco_ratio	Locomotion-independent exploration into the internal boxes during the 300s: The ratio of internal visits to locomotor activity during the 300s	Ratio	
Expl_E_I_TOT_Loco_ratio	Locomotion-independent exploration into the boxes during the 300s: The ratio of visits to locomotor activity during the 300s	Ratio	
LAT_E	Latency of the first exploration into an external box.	ms	
LAT_I	Latency of the first exploration into an internal box.	ms	
LAT_E_I	Latency of the first exploration into a box.	ms	
Lat_Tot_Log	Logarithmic value of the latency of the first exploration into a box.	Log(ms)	
Lat_E_Log	Logarithmic value of the latency of the first exploration into an external box.	Log(ms)	
Lat_I_Log	Logarithmic value of the latency of the first exploration into an internal box.	Log(ms)	
Box_E_BEF_Nr	Number of visited external boxes up to EAT_TOT_T	Number (int or float)	Maximum:8
Box_I_BEF_Nr	Number of visited internal boxes up to EAT_TOT_T	Number (int or float)	Maximum:8
Box_E_I_BEF_Nr	Number of visited boxes up to EAT_TOT_T	Number (int or float)	Maximum:16
Box_REP_E_BEF_Nr	Number of repeatedly visited external boxes up to EAT_TOT_T	Number (int or float)	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Box_REP_I_BEF_Nr	Number of repeatedly visited internal boxes up to EAT_TOT_T	Number (int or float)	
Box_REP_E_I_BEF_Nr	Number of repeatedly visited boxes up to EAT_TOT_T	Number (int or float)	
Expl_REP_E_BEF_Nr	Number of repetition of the explorations into the external boxes up to EAT_TOT_T	Number (int or float)	
Expl_REP_I_BEF_Nr	Number of repetition of the explorations into the internal boxes up to EAT_TOT_T	Number (int or float)	
Expl_REP_BEF_Nr	Number of repetition of the explorations into boxes up to EAT_TOT_T	Number (int or float)	
Expl_REP_E_BEF_Calc_T	Exploration repetitions in the external boxes up to collection of the rewards (Number of external box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_REP_I_BEF_Calc_T	Exploration repetitions in the internal boxes up to collection of the rewards (Number of internal box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_REP_E_I_BEF_Calc_T	Exploration repetitions in the all boxes up to collection of the rewards (Number of box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_REP_E_AFT_Nr	Number of repetition of the explorations into the external boxes after EAT_TOT_T	Number (int or float)	If the eating time is 300, there is no value
Expl_REP_I_AFT_Nr	Number of repetition of the explorations into the internal boxes after EAT_TOT_T	Number (int or float)	If the eating time is 300, there is no value
Expl_REP_AFT_Nr	Number of repetition of the explorations into boxes after EAT_TOT_T	Number (int or float)	
Expl_REP_E_AFT_Nr_Calc	Exploration repetitions in the external boxes after the collection of the rewards (Number of external box visit repetition $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Expl_REP_I_AFT_Nr_Calc	Exploration repetitions in the internal boxes after the collection of the rewards (Number of internal box visit repetition $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	
Expl_REP_AFT_Nr_Calc	Exploration repetitions in boxes after the collection of the rewards (Number of box visit repetition $\times$ 300) / (300-EAT_TOT_T)	Number (int or float)	
Expl_REP_E_BEF_AFT_Nr	Number of repeated external box exploration during 300s	Number (int or float)	
Expl_REP_I_BEF_AFT_Nr	Number of repeated internal box exploration during 300s	Number (int or float)	
Expl_REP_TOT_Nr	Number of repeated all box exploration during 300s	Number (int or float)	
Learning related parameters			
L_C_E	Learning capacity at the external boxes: (EAT_E_Nr x 300 x 100) / [(number of available rewards; [8]) x (EAT_E_T)]	%	No food there, no value (Trial 3,4)
L_C_I	Learning capacity at the internal boxes: (EAT_I_Nr x 300 x 100) / [(number of available rewards; [8]) x (EAT_I_T)]	%	There is value for 1-4 trials
L_C_Tot	Learning capacity at both sides: (EAT_TOT_Nr x 300 x 100) / [(number of rewards) x (EAT_TOT_T)]	%	Only trial 1-2
L_C	Learning capacity for all trials at both sides: (EAT_TOT_Nr x 300 x 100) / [(number of rewards) x (EAT_TOT_T)]	%	
A_E	Adequate exploration: (EAT_TOT_Nr x 100) / (number of explorations up to EAT_TOT_T)	%	If the divisor is 0 the value is 0
Error related parameters			
First_Box location	The side of the first visit (1=internal; 2=external box)	Number (int or float)	Discrete feature only value 1 or 2
Skipping	The number of entries into the corridors up to the first exploration	Number (int or float)	

Feature desription for LE_Lisket_0_15_AMBI_DEF_2.xlsx - continued

Feature name	Description	Unit	Note
Expl_E_BEF_Ratio	The ratio of visits to external and total explorations up to collection of all rewards	ratio	
Expl_E_TOT_Ratio	The ratio of visits to external and total explorations during 300s	ratio	
Eff_Expl_E	(EAT_E_Nr x 100) / (number of external box visited [max:8])	ratio	Only trial 1-2
Eff_Expl_I	(EAT_I_Nr x 100) / (number of internal box visited [max:8])	%	
Eff_Expl_EI	(EAT_TOT_Nr x 100) / (number of box visited [max:16])	%	Only trial 1-2
E_E	Effective exploration. (EAT_TOT_Nr x 100) / (number of box visited [max:16 for trials 1-2 and max:8 for trials 3-4])	%	

Table 2: Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv

Feature Name	Description	Unit	Note
Generation	Number of the generation where the rat belongs	Number (int or float)	
Season	The season when the animals were delivered	Text (str)	
Paradigm	Types of tasks: Task1 or Task2	Text (str)	
Year	Year of the experiment (0-6)	Number (int or float)	
NR	ID number of the animal	Number (int or float)	
Group	The group label of the rat	Text (str)	
Sex	Sex of the rat	Text (str)	
EAT_E_%	The percentage of the collected rewards from the external boxes (number of collected rewards x 100) / (available rewards [8])	%	At first trial
EAT_I_%	The percentage of the collected rewards from the internal boxes (number of collected rewards x 100) / (available rewards [8])	%	At first trial
EAT_TOT_%%	The percentage of the collected rewards from all boxes (number of collected rewards x 100) / (available rewards [16])	%	At first trial
EAT_E_T	The time up to the collection of the rewards from the external boxes (cut off time: 300s)	Second	At first trial
EAT_I_T	The time up to the collection of the rewards from the internal boxes (cut off time: 300s)	Second	At first trial
EAT_TOT_T	The time up to the collection of the rewards from all boxes (cut off time: 300s)	Second	At first trial
Expl_E_BEF_Calc	Subset exploration categorized as external box visits up to collection of the rewards (Number of external box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	
Expl_I_BEF_Calc	Subset of exploration categorized as internal box visits up to collection of the rewards (Number of internal box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
EXPL_BEF	Subset of exploration up to collection of the rewards (Number of box exploration up to collection of reward $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial
Expl_E_BEF_AFT_Nr	Number of external box exploration during 300s	Number (int or float)	At first trial
Expl_I_BEF_AFT_Nr	Number of internal box exploration during 300s	Number (int or float)	At first trial
EXPL_TOT	Number of all box exploration during 300s	Number (int or float)	At first trial
Expl_E_BEF_AFT_T	Duration of external box exploration during 300s	Second	At first trial
Expl_I_BEF_AFT_T	Duration of internal box exploration during 300s	Second	At first trial
Expl_TOT_T	Duration of all box exploration during 300s	Second	At first trial
Lat_Tot_Log	Logarithmic value of the latency of the first exploration into a box.	Log(ms)	At first trial
Lat_E_Log	Logarithmic value of the latency of the first exploration into an external box.	Log(ms)	At first trial
Lat_I_Log	Logarithmic value of the latency of the first exploration into an internal box.	Log(ms)	At first trial
Box_E_BEF_Nr	Number of visited external boxes up to EAT_TOT_T	Number (int or float)	At first trial
Box_I_BEF_Nr	Number of visited internal boxes up to EAT_TOT_T	Number (int or float)	At first trial
Box_E_I_BEF_Nr	Number of visited boxes up to EAT_TOT_T	Number (int or float)	At first trial
Box_REP_E_BEF_Nr	Number of repeatedly visited external boxes up to EAT_TOT_T	Number (int or float)	At first trial
Box_REP_I_BEF_Nr	Number of repeatedly visited internal boxes up to EAT_TOT_T	Number (int or float)	At first trial
Box_REP_E_I_BEF_Nr	Number of repeatedly visited boxes up to EAT_TOT_T	Number (int or float)	At first trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_REP_E_BEF_Calc_T	Exploration repetitions in the external boxes up to collection of the rewards (Number of external box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial
Expl_REP_I_BEF_Calc_T	Exploration repetitions in the internal boxes up to collection of the rewards (Number of internal box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial
Expl_REP_E_I_BEF_Calc_T	Exploration repetitions in the all boxes up to collection of the rewards (Number of box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial
Expl_REP_E_BEF_AFT_Nr	Number of repeated external box exploration during 300s	Number (int or float)	At first trial
Expl_REP_I_BEF_AFT_Nr	Number of repeated internal box exploration during 300s	Number (int or float)	At first trial
Expl_REP_TOT_Nr	Number of repeated all box exploration during 300s	Number (int or float)	At first trial
Dir_Change_Nr	The number of direction changes during the 300s	Number (int or float)	At first trial
Circle_Nr	The number of total circles during the 300s	Number (int or float)	At first trial
LOCO_TOT	The number of entries into the corridors during the 300s	Number (int or float)	At first trial
Explored Area	The number of visited corridors (max:4)	Number (int or float)	At first trial
Loco_BEF_NR	The number of entries into the corridors up to collection of rewards (Number of corridor entries up to collection of rewards $\times$ 300) / EAT_TOT_T	Number (int or float)	At first trial
Expl_E_BEF_Ratio	The ratio of visits to external and total explorations up to collection of all rewards	ratio	At first trial
Expl_E_TOT_Ratio	The ratio of visits to external and total explorations during 300s	ratio	At first trial
L_C_E	Learning capacity at the external boxes: (EAT_E_Nr $\times$ 300 $\times$ 100) / [(number of available rewards; [8]) $\times$ (Eat_E_T)]	%	At first trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
L_C_I	Learning capacity at the internal boxes: $(EAT_I_Nr \times 300 \times 100) / [(number\ of\ available\ rewards; [8]) \times (EAT_I_T)]$	%	At first trial
L_C	Learning capacity for all trials at both sides: $(EAT_TOT_Nr \times 300 \times 100) / [(number\ of\ rewards) \times (EAT_TOT_T)]$	%	At first trial
A_E	Adequate exploration: $(EAT_TOT_Nr \times 100) / (number\ of\ explorations\ up\ to\ EAT_TOT_T)$	%	At first trial
Expl_E_BEF_Loco_ratio	Locomotion-independent exploration into the external boxes up to the EAT_TOT_T: The ratio of external visits to locomotor activity until all rewards were collected	Number (int or float)	At first trial
Expl_I_BEF_Loco_ratio	Locomotion-independent exploration into the internal boxes up to the EAT_TOT_T: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At first trial
Expl_EI_BEF_Loco_ratio	Locomotion-independent exploration into the boxes up to the EAT_TOT_T: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At first trial
Expl_E_TOT_Loco_ratio	Locomotion-independent exploration into the external boxes during the 300s: The ratio of external visits to locomotor activity until all rewards were collected	Ratio	At first trial
Expl_I_TOT_Loco_ratio	Locomotion-independent exploration into the internal boxes during the 300s: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At first trial
Expl_E_I_TOT_Loco_ratio	Locomotion-independent exploration into the boxes during the 300s: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At first trial
Eff_Expl_E	$(EAT_E_Nr \times 100) / (number\ of\ external\ box\ visited [max:8])$	ratio	At first trial
Eff_Expl_I	$(EAT_I_Nr \times 100) / (number\ of\ internal\ box\ visited [max:8])$	%	At first trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Eff_Expl_EI	$(\text{EAT_TOT_Nr} \times 100) / (\text{number of box visited } [\text{max:16}])$	%	At first trial
E_E	Effective exploration. $(\text{EAT_TOT_Nr} \times 100) / (\text{number of box visited } [\text{max:16 for trials 1-2 and max:8 for trials 3-4}])$	%	At first trial
id	Unique identifier constructed from Group, NR and Generation	Text (str)	
EAT_E_%_2	The percentage of the collected rewards from the external boxes (number of collected rewards $\times 100$) / (available rewards [8])	%	At second trial
EAT_I_%_2	The percentage of the collected rewards from the internal boxes (number of collected rewards $\times 100$) / (available rewards [8])	%	At second trial
EAT_TOT_%%_2	The percentage of the collected rewards from all boxes (number of collected rewards $\times 100$) / (available rewards [16])	%	At second trial
EAT_E_T_2	The time up to the collection of the rewards from the external boxes (cut off time: 300s)	Second	At second trial
EAT_I_T_2	The time up to the collection of the rewards from the internal boxes (cut off time: 300s)	Second	At second trial
EAT_TOT_T_2	The time up to the collection of the rewards from all boxes (cut off time: 300s)	Second	At second trial
Expl_E_BEF_Calc_2	Subset exploration categorized as external box visits up to collection of the rewards (Number of external box exploration $\times 300$) / EAT_TOT_T	Number (int or float)	At second trial
Expl_I_BEF_Calc_2	Subset of exploration categorized as internal box visits up to collection of the rewards (Number of internal box exploration $\times 300$) / EAT_TOT_T	Number (int or float)	At second trial
EXPL_BEF_2	Subset of exploration up to collection of the rewards (Number of box exploration up to collection of reward $\times 300$) / EAT_TOT_T	Number (int or float)	At second trial
Expl_E_BEF_AFT_Nr_2	Number of external box exploration during 300s	Number (int or float)	At second trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_I_BEF_AFT_Nr_2	Number of internal box exploration during 300s	Number (int or float)	At second trial
EXPL_TOT_2	Number of all box exploration during 300s	Number (int or float)	At second trial
Expl_E_BEF_AFT_T_2	Duration of external box exploration during 300s	Second	At second trial
Expl_I_BEF_AFT_T_2	Duration of internal box exploration during 300s	Second	At second trial
Expl_TOT_T_2	Duration of all box exploration during 300s	Second	At second trial
Lat_Tot_Log_2	Logarithmic value of the latency of the first exploration into an external box.	Log(ms)	At second trial
Lat_E_Log_2	Logarithmic value of the latency of the first exploration into an internal box.	Log(ms)	At second trial
Lat_I_Log_2	Logarithmic value of the latency of the first exploration into a box.	Log(ms)	At second trial
Box_E_BEF_Nr_2	Number of visited external boxes up to EAT_TOT_T	Number (int or float)	At second trial
Box_I_BEF_Nr_2	Number of visited internal boxes up to EAT_TOT_T	Number (int or float)	At second trial
Box_E_I_BEF_Nr_2	Number of visited boxes up to EAT_TOT_T	Number (int or float)	At second trial
Box_REP_E_BEF_Nr_2	Number of repeatedly visited external boxes up to EAT_TOT_T	Number (int or float)	At second trial
Box_REP_I_BEF_Nr_2	Number of repeatedly visited internal boxes up to EAT_TOT_T	Number (int or float)	At second trial
Box_REP_E_I_BEF_Nr_2	Number of repeatedly visited boxes up to EAT_TOT_T	Number (int or float)	At second trial
Expl_REP_E_BEF_Calc_T_2	Exploration repetitions in the external boxes up to collection of the rewards (Number of external box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At second trial
Expl_REP_I_BEF_Calc_T_2	Exploration repetitions in the internal boxes up to collection of the rewards (Number of internal box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At second trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_REP_E_I_BEF_Calc_T_2	Exploration repetitions in the all boxes up to collection of the rewards (Number of box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At second trial
Expl_REP_E_BEF_AFT_Nr_2	Number of repeated external box exploration during 300s	Number (int or float)	At second trial
Expl_REP_I_BEF_AFT_Nr_2	Number of repeated internal box exploration during 300s	Number (int or float)	At second trial
Expl_REP_TOT_Nr_2	Number of repeated all box exploration during 300s	Number (int or float)	At second trial
Dir_Change_Nr_2	The number of direction changes during the 300s	Number (int or float)	At second trial
Circle_Nr_2	The number of total circles during the 300s	Number (int or float)	At second trial
LOCO_TOT_2	The number of entries into the corridors during the 300s	Number (int or float)	At second trial
Explored Area_2	The number of visited corridors (max:4)	Number (int or float)	At second trial
Expl_E_BEF_Ratio_2	The ratio of visits to external and total explorations up to collection of all rewards	ratio	At second trial
Expl_E_TOT_Ratio_2	The ratio of visits to external and total explorations during 300s	ratio	At second trial
L_C_E_2	Learning capacity at the external boxes: (EAT_E_Nr $\times$ 300 $\times$ 100) / [(number of available rewards; [8]) $\times$ (Eat_E_T)]	%	At second trial
L_C_I_2	Learning capacity at the internal boxes: (EAT_I_Nr $\times$ 300 $\times$ 100) / [(number of available rewards; [8]) $\times$ (EAT_I_T)]	%	At second trial
L_C_2	Learning capacity for all trials at both sides: (EAT_TOT_Nr $\times$ 300 $\times$ 100) / [(number of rewards) $\times$ (EAT_TOT_T)]	%	At second trial
A_E_2	Adequate exploration: (EAT_TOT_Nr $\times$ 100) / (number of explorations up to EAT_TOT_T)	%	At second trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_E_BEF_Loco_ratio_2	Locomotion-independent exploration into the external boxes up to the EAT_TOT_T: The ratio of external visits to locomotor activity until all rewards were collected	Number (int or float)	At second trial
Expl_I_BEF_Loco_ratio_2	Locomotion-independent exploration into the internal boxes up to the EAT_TOT_T: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At second trial
Expl_EI_BEF_Loco_ratio_2	Locomotion-independent exploration into the boxes up to the EAT_TOT_T: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At second trial
Expl_E_TOT_Loco_ratio_2	Locomotion-independent exploration into the external boxes during the 300s: The ratio of external visits to locomotor activity until all rewards were collected	Ratio	At second trial
Expl_I_TOT_Loco_ratio_2	Locomotion-independent exploration into the internal boxes during the 300s: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At second trial
Expl_E_I_TOT_Loco_ratio_2	Locomotion-independent exploration into the boxes during the 300s: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At second trial
Eff_Expl_E_2	(EAT_E_Nr x 100) / (number of external box visited [max:8])	ratio	At second trial
Eff_Expl_I_2	(EAT_I_Nr x 100) / (number of internal box visited [max:8])	%	At second trial
Eff_Expl_EI_2	(EAT_TOT_Nr x 100) / (number of box visited [max:16])	%	At second trial
E_E_2	Effective exploration. (EAT_TOT_Nr x 100) / (number of box visited [max:16 for trials 1-2 and max:8 for trials 3-4])	%	At second trial
EAT_I_%_3	The percentage of the collected rewards from the internal boxes (number of collected rewards x 100) / (available rewards [8])	%	At third trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
EAT_TOT_%%_3	The percentage of the collected rewards from all boxes (number of collected rewards x 100) / (available rewards [16])	%	At third trial
EAT_I_T_3	The time up to the collection of the rewards from the internal boxes (cut off time: 300s)	Second	At third trial
EAT_TOT_T_3	The time up to the collection of the rewards from all boxes (cut off time: 300s)	Second	At third trial
Expl_E_BEF_Calc_3	Subset exploration categorized as external box visits up to collection of the rewards (Number of external box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	At third trial
Expl_I_BEF_Calc_3	Subset of exploration categorized as internal box visits up to collection of the rewards (Number of internal box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	At third trial
EXPL_BEF_3	Subset of exploration up to collection of the rewards (Number of box exploration up to collection of reward $\times$ 300) / EAT_TOT_T	Number (int or float)	At third trial
Expl_E_BEF_AFT_Nr_3	Number of external box exploration during 300s	Number (int or float)	At third trial
Expl_I_BEF_AFT_Nr_3	Number of internal box exploration during 300s	Number (int or float)	At third trial
EXPL_TOT_3	Number of all box exploration during 300s	Number (int or float)	At third trial
Expl_E_BEF_AFT_T_3	Duration of external box exploration during 300s	Second	At third trial
Expl_I_BEF_AFT_T_3	Duration of internal box exploration during 300s	Second	At third trial
Expl_TOT_T_3	Duration of all box exploration during 300s	Second	At third trial
Lat_Tot_Log_3	Logarithmic value of the latency of the first exploration into an external box.	Log(ms)	At third trial
Lat_E_Log_3	Logarithmic value of the latency of the first exploration into an internal box.	Log(ms)	At third trial
Lat_I_Log_3	Logarithmic value of the latency of the first exploration into a box.	Log(ms)	At third trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Box_E_BEF_Nr_3	Number of visited external boxes up to EAT_TOT_T	Number (int or float)	At third trial
Box_I_BEF_Nr_3	Number of visited internal boxes up to EAT_TOT_T	Number (int or float)	At third trial
Box_E_I_BEF_Nr_3	Number of visited boxes up to EAT_TOT_T	Number (int or float)	At third trial
Box_REP_E_BEF_Nr_3	Number of repeatedly visited external boxes up to EAT_TOT_T	Number (int or float)	At third trial
Box_REP_I_BEF_Nr_3	Number of repeatedly visited internal boxes up to EAT_TOT_T	Number (int or float)	At third trial
Box_REP_E_I_BEF_Nr_3	Number of repeatedly visited boxes up to EAT_TOT_T	Number (int or float)	At third trial
Expl_REP_E_BEF_Calc_T_3	Exploration repetitions in the external boxes up to collection of the rewards ($\text{Number of external box visit repetition} \times 300$) / EAT_TOT_T	Number (int or float)	At third trial
Expl_REP_I_BEF_Calc_T_3	Exploration repetitions in the internal boxes up to collection of the rewards ($\text{Number of internal box visit repetition} \times 300$) / EAT_TOT_T	Number (int or float)	At third trial
Expl_REP_E_I_BEF_Calc_T_3	Exploration repetitions in the all boxes up to collection of the rewards ($\text{Number of box visit repetition} \times 300$) / EAT_TOT_T	Number (int or float)	At third trial
Expl_REP_E_BEF_AFT_Nr_3	Number of repeated external box exploration during 300s	Number (int or float)	At third trial
Expl_REP_I_BEF_AFT_Nr_3	Number of repeated internal box exploration during 300s	Number (int or float)	At third trial
Expl_REP_TOT_Nr_3	Number of repeated all box exploration during 300s	Number (int or float)	At third trial
Dir_Change_Nr_3	The number of direction changes during the 300s	Number (int or float)	At third trial
Circle_Nr_3	The number of total circles during the 300s	Number (int or float)	At third trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
LOCO_TOT_3	The number of entries into the corridors during the 300s	Number (int or float)	At third trial
Explored Area_3	The number of visited corridors (max:4)	Number (int or float)	At third trial
Expl_E_BEF_Ratio_3	The ratio of visits to external and total explorations up to collection of all rewards	ratio	At third trial
Expl_E_TOT_Ratio_3	The ratio of visits to external and total explorations during 300s	ratio	At third trial
L_C_I_3	Learning capacity at the internal boxes: $(EAT_I_Nr \times 300 \times 100) / [(number\ of\ available\ rewards; [8]) \times (EAT_I_T)]$	%	At third trial
L_C_3	Learning capacity for all trials at both sides: $(EAT_TOT_Nr \times 300 \times 100) / [(number\ of\ rewards) \times (EAT_TOT_T)]$	%	At third trial
A_E_3	Adequate exploration: $(EAT_TOT_Nr \times 100) / (number\ of\ explorations\ up\ to\ EAT_TOT_T)$	%	At third trial
Expl_E_BEF_Loco_ratio_3	Locomotion-independent exploration into the external boxes up to the EAT_TOT_T: The ratio of external visits to locomotor activity until all rewards were collected	Number (int or float)	At third trial
Expl_I_BEF_Loco_ratio_3	Locomotion-independent exploration into the internal boxes up to the EAT_TOT_T: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At third trial
Expl_EI_BEF_Loco_ratio_3	Locomotion-independent exploration into the boxes up to the EAT_TOT_T: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At third trial
Expl_E_TOT_Loco_ratio_3	Locomotion-independent exploration into the external boxes during the 300s: The ratio of external visits to locomotor activity until all rewards were collected	Ratio	At third trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_I_TOT_Loco_ratio_3	Locomotion-independent exploration into the internal boxes during the 300s: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At third trial
Expl_E_I_TOT_Loco_ratio_3	Locomotion-independent exploration into the boxes during the 300s: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At third trial
Eff_Expl_I_3	$(EAT_I_Nr \times 100) / (\text{number of internal box visited} [\text{max:8}])$	%	At third trial
Eff_Expl_EI_3	$(EAT_TOT_Nr \times 100) / (\text{number of box visited} [\text{max:16}])$	%	At third trial
E_E_3	Effective exploration. $(EAT_TOT_Nr \times 100) / (\text{number of box visited} [\text{max:16 for trials 1-2 and max:8 for trials 3-4}])$	%	At third trial
EAT_I_%_4	The percentage of the collected rewards from the internal boxes (number of collected rewards x 100) / (available rewards [8])	%	At fourth trial
EAT_TOT_%%_4	The percentage of the collected rewards from all boxes (number of collected rewards x 100) / (available rewards [16])	%	At fourth trial
EAT_I_T_4	The time up to the collection of the rewards from the internal boxes (cut off time: 300s)	Second	At fourth trial
EAT_TOT_T_4	The time up to the collection of the rewards from all boxes (cut off time: 300s)	Second	At fourth trial
Expl_E_BEF_Calc_4	Subset exploration categorized as external box visits up to collection of the rewards (Number of external box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial
Expl_I_BEF_Calc_4	Subset of exploration categorized as internal box visits up to collection of the rewards (Number of internal box exploration $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial
EXPL_BEF_4	Subset of exploration up to collection of the rewards (Number of box exploration up to collection of reward $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_E_BEF_AFT_Nr_4	Number of external box exploration during 300s	Number (int or float)	At fourth trial
Expl_I_BEF_AFT_Nr_4	Number of internal box exploration during 300s	Number (int or float)	At fourth trial
EXPL_TOT_4	Number of all box exploration during 300s	Number (int or float)	At fourth trial
Expl_E_BEF_AFT_T_4	Duration of external box exploration during 300s	Second	At fourth trial
Expl_I_BEF_AFT_T_4	Duration of internal box exploration during 300s	Second	At fourth trial
Expl_TOT_T_4	Duration of all box exploration during 300s	Second	At fourth trial
Lat_Tot_Log_4	Logarithmic value of the latency of the first exploration into a box.	Log(ms)	At fourth trial
Lat_E_Log_4	Logarithmic value of the latency of the first exploration into an external box.	Log(ms)	At fourth trial
Lat_I_Log_4	Logarithmic value of the latency of the first exploration into an internal box.	Log(ms)	At fourth trial
Box_E_BEF_Nr_4	Number of visited external boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Box_I_BEF_Nr_4	Number of visited internal boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Box_E_I_BEF_Nr_4	Number of visited boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Box_REP_E_BEF_Nr_4	Number of repeatedly visited external boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Box_REP_I_BEF_Nr_4	Number of repeatedly visited internal boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Box_REP_E_I_BEF_Nr_4	Number of repeatedly visited boxes up to EAT_TOT_T	Number (int or float)	At fourth trial
Expl_REP_E_BEF_Calc_T_4	Exploration repetitions in the external boxes up to collection of the rewards (Number of external box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_REP_I_BEF_Calc_T_4	Exploration repetitions in the internal boxes up to collection of the rewards (Number of internal box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial
Expl_REP_E_I_BEF_Calc_T_4	Exploration repetitions in the all boxes up to collection of the rewards (Number of box visit repetition $\times$ 300) / EAT_TOT_T	Number (int or float)	At fourth trial
Expl_REP_E_BEF_AFT_Nr_4	Number of repeated external box exploration during 300s	Number (int or float)	At fourth trial
Expl_REP_I_BEF_AFT_Nr_4	Number of repeated internal box exploration during 300s	Number (int or float)	At fourth trial
Expl_REP_TOT_Nr_4	Number of repeated all box exploration during 300s	Number (int or float)	At fourth trial
Dir_Change_Nr_4	The number of direction changes during the 300s	Number (int or float)	At fourth trial
Circle_Nr_4	The number of total circles during the 300s	Number (int or float)	At fourth trial
LOCO_TOT_4	The number of entries into the corridors during the 300s	Number (int or float)	At fourth trial
Explored Area_4	The number of visited corridors (max:4)	Number (int or float)	At fourth trial
Expl_E_BEF_Ratio_4	The ratio of visits to external and total explorations up to collection of all rewards	ratio	At fourth trial
Expl_E_TOT_Ratio_4	The ratio of visits to external and total explorations during 300s	ratio	At fourth trial
L_C_I_4	Learning capacity at the internal boxes: (EAT_I_Nr $\times$ 300 $\times$ 100) / [(number of available rewards; 8] $\times$ (EAT_I_T)]	%	At fourth trial
L_C_4	Learning capacity for all trials at both sides: (EAT_TOT_Nr $\times$ 300 $\times$ 100) / [(number of rewards) $\times$ (EAT_TOT_T)]	%	At fourth trial
A_E_4	Adequate exploration: (EAT_TOT_Nr $\times$ 100) / (number of explorations up to EAT_TOT_T)	%	At fourth trial

Feature descriptions for ambitus_0_15_ml_ready_04_08_2025.csv – continued

Feature Name	Description	Unit	Note
Expl_E_BEF_Loco_ratio_4	Locomotion-independent exploration into the external boxes up to the EAT_TOT_T: The ratio of external visits to locomotor activity until all rewards were collected	Number (int or float)	At fourth trial
Expl_I_BEF_Loco_ratio_4	Locomotion-independent exploration into the internal boxes up to the EAT_TOT_T: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At fourth trial
Expl_EI_BEF_Loco_ratio_4	Locomotion-independent exploration into the boxes up to the EAT_TOT_T: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At fourth trial
Expl_E_TOT_Loco_ratio_4	Locomotion-independent exploration into the external boxes during the 300s: The ratio of external visits to locomotor activity until all rewards were collected	Ratio	At fourth trial
Expl_I_TOT_Loco_ratio_4	Locomotion-independent exploration into the internal boxes during the 300s: The ratio of internal visits to locomotor activity until all rewards were collected	Ratio	At fourth trial
Expl_E_I_TOT_Loco_ratio_4	Locomotion-independent exploration into the boxes during the 300s: The ratio of visits to locomotor activity until all rewards were collected	Ratio	At fourth trial
Eff_Expl_I_4	(EAT_I_Nr x 100) / (number of internal box visited [max:8])	%	At fourth trial
Eff_Expl_EI_4	(EAT_TOT_Nr x 100) / (number of box visited [max:16])	%	At fourth trial
E_E_4	Effective exploration. (EAT_TOT_Nr x 100) / (number of box visited [max:16 for trials 1-2 and max:8 for trials 3-4])	%	At fourth trial

Table 3: Published files detailed description

Name	Type	Dimensions	Size
ambitus_0_15_log_04_08_2025	Parquet	5368 x 105	1.3MB
ambitus_0_15_ml_ready_04_08_2025	CSV	1,342 x 202	2.5MB